

POWER-TOOL BATTERY CELL - DELIVERY PACKAGE

Virtual Battery Factory / case t3_r3 / champion V12-HiP-final
Nominal capacity ≥ 2 Ah / 5C retention $\geq 95\%$ / 4C fast charge (no plating) / $T_{\text{max}} \leq 60$ C /
power density ≥ 4000 W/kg

Case name	t3_r3 - power-tool battery	
Numbering scheme	VBF-<CASE-ID-UPPERCASE>-<DOC-CODE>-<SEQ-NO> (case ID t3_r3 -> T3R3)	
Generation date	2026-08-26	
Prepared:	Reviewed:	Approved:
(left blank for manual signing)		

Document Code Reference (fixed by protocol)

Document code	Meaning	Corresponding file
DS	Specification	design_spec.md
BOM	Bill of Materials	bom.xlsx
DSH	Datasheet technical parameter sheet	datasheet.docx
CALC	Calculation sheet	calc.xlsx
DVPR	Design verification report	dvpr.md
DFMEA	Failure analysis	dfmea.md
CAD	Structure model	cell_model.stl (not generated: no 3D structure model requested this case)

File List

File name	Number	Format	Source description
design_spec.md	VBF-T3R3-DS-01	md	Cell design specification generated per deliverable-design-spec format (build script, values from tool outputs)
design_spec.pdf	VBF-T3R3-DS-01	pdf	PDF release of design_spec.md, rendered with reportlab
report.html	VBF-T3R3-DS-02	html	Case report deterministically rendered from log.jsonl (bda render --case-dir runs/exp/t3_r3); belongs to DS
bom.xlsx	VBF-T3R3-BOM-01	xlsx	Bill of Materials, dual caliber g/cell and kg/kWh (openpyxl; formulas and sources per row)
bom.pdf	VBF-T3R3-BOM-01	pdf	PDF release of bom.xlsx, sheets rendered with reportlab
datasheet.docx	VBF-T3R3-DSH-01	docx	Customer-facing technical datasheet (python-docx; per-line sources)
datasheet.pdf	VBF-T3R3-DSH-01	pdf	PDF release of datasheet.docx, rendered with reportlab
calc.xlsx	VBF-T3R3-CALC-01	xlsx	Design calculation sheet: inputs -> capacity and energy -> energy density -> N/P and mass -> process parameters (openpyxl)
calc.pdf	VBF-T3R3-CALC-01	pdf	PDF release of calc.xlsx, sheets rendered with reportlab
dvpr.md	VBF-T3R3-DVPR-01	md	Design Verification Plan and Report, virtual test version (simulation protocols)
dvpr.pdf	VBF-T3R3-DVPR-01	pdf	PDF release of dvpr.md, rendered with reportlab
dfmea.md	VBF-T3R3-DFMEA-01	md	Design FMEA, qualitative version based on simulation risk signals
dfmea.pdf	VBF-T3R3-DFMEA-01	pdf	PDF release of dfmea.md, rendered with reportlab
delivery_index.md	VBF-T3R3-IDX-01	md	Delivery package index: cover + controlled file list (this file)
delivery_index.pdf	VBF-T3R3-IDX-01	pdf	PDF release of delivery_index.md, reportlab with blueprint colors